Lecture Scribe Request

generate a lecture scribe intended to serve as exam reference material for students. The scribe must faithfully reconstruct the content of the provided lecture slides and relevant textbook material only.

While writing the scribe, reason step by step through all definitions, assumptions, theorem statements, proofs, derivations, and worked examples exactly as presented in the provided context. The logical flow between statements and steps must be made explicit so that a student can follow the reasoning during revision.

Include:

Definitions and notation

Assumptions and conditions

Statements of theorems or results

Step-by-step proofs or derivations when present in the lecture

Step-by-step worked examples only if they appear in the provided material

Do not introduce new explanations, intuition, examples, alternative methods, or results that are not present in the lecture or textbook context. Maintain a formal academic tone suitable for exam preparation.

The structure of the scribe should follow the organization and scope of the lecture.

I can definitely generate that kind of **exam-ready lecture scribe**, but I need the **actual lecture content first**.

Please upload or provide one of the following:

- 📄 Lecture slides (PDF / images)
- 📘 Relevant textbook pages
- 📝 Notes or screenshots

Once you share the material, I will:

- Reconstruct everything **faithfully (no extra content)**
- Write **step-by-step derivations and proofs**
- Keep **formal academic tone**
- Organize it exactly like your lecture
- Make it **perfect for revision before exams**

Upload the content and I'll start immediately.

L15_L17.pdf
PDF

Here is your **formal lecture scribe (exam-ready)** reconstructed strictly from the uploaded slides:

CSE400 – Fundamentals of Probability in Computing

Lecture L15–L18: Joint Random Variables and Distributions

---

1. Motivation: Why Study Joint Random Variables

In many real-life and scientific applications, we are interested in **two or more random variables simultaneously**.

Examples (from lecture)

- Height and weight of giraffes
- IQ and birthweight of children
- Frequency of exercise and heart disease rate
- Air pollution level and respiratory illness rate
- Number of Facebook friends and age of members

Structured Examples

1. **Smart Transportation**
   - $X =$ Vehicle Speed
   - $Y =$ Traffic Density
   - Question: Can both be high simultaneously?
   - Conclusion: Variables are **correlated**
2. **Wireless Networks**
   - $X =$ Signal-to-Noise Ratio (SNR)
   - $Y =$ Throughput
   - Throughput depends on SNR but not deterministically
3. **Weather Prediction**
   - $X =$ Rainfall, $Y =$ Temperature
4. **Drone Delivery**
   - $X =$ Wind Speed, $Y =$ Delivery Time
5. **Two Dice Example**
   - Joint behavior required for conditions such as $X + Y = 7$

Key Conclusion

- Random variables have a **joint distribution** that:
  - Allows computation of probabilities involving both variables
  - Helps understand **relationships** between variables
- If variables are not independent, dependence is measured using:
  - Covariance
  - Correlation

---

2. Discrete Case – Joint PMF

Setup

Let:

- $X = \{x_1, x_2, \ldots, x_n\}$
- $Y = \{y_1, y_2, \ldots, y_m\}$

Then the pair:

$$(X, Y) \in \{(x_1, y_1), (x_1, y_2), \ldots, (x_n, y_m)\}$$

Definition (Joint PMF)

$$p(x_i, y_j) = P(X = x_i, Y = y_j)$$

This represents the **probability of the joint outcome occurring simultaneously**.

Joint PMF Table Representation

The joint PMF is organized in a table with:

- Rows → values of X
- Columns → values of Y
- Entries → $p(x_i, y_j)$

Key Properties

1.

0 \le p(x_i, y_j) \le 1

]

2.

\sum_{i=1}^{n} \sum_{j=1}^{m} p(x_i, y_j) = 1

]

---

3. Example: Rolling Two Dice

Experiment

- Roll two fair dice

Let:

- $X =$ value on first die
- $T =$ total (sum) of both dice

Joint Distribution Construction

- Each outcome (i, j) has probability:

$$\frac{1}{36}$$

- Construct joint table of (X, T)

Observation (from table)

- Entries are either 0 or $\frac{1}{36}$
- X and T are **not independent**

---

4. Continuous Case – Joint PDF

Analogy with Discrete Case

- Discrete values → Continuous intervals
- Joint PMF → Joint PDF
- Summation → Double integration

Domain

If:

- $X \in [a, b]$, $Y \in [c, d]$

Then:

$$(X, Y) \in [a, b] \times [c, d]$$

Definition (Joint PDF)

A function $f(x, y)$ is a joint PDF if:

$$P((X, Y) \text{ in small rectangle}) \approx f(x, y)\, dx\, dy$$

Key Properties

1.

f(x,y) \ge 0

]

2.

\int_c^d \int_a^b f(x,y),dx,dy = 1

]

Important Note

- $f(x, y)$ may be greater than 1
- It is a **density**, not a probability

---

5. Worked Example (Continuous Case)

Given

$$f(x, y) = 4xy, \quad 0 \le x \le 1,\ 0 \le y \le 1$$

Tasks

1. Show it is a valid joint PDF
2. Visualize event:

$$A = \{X < 0.5,\ Y > 0.5\}$$

3. Compute $P(A)$

Step 1: Check Validity (Normalization)

$$\int_0^1 \int_0^1 4xy\, dx\, dy = \int_0^1 \left(\int_0^1 4xy\, dx \right) dy$$

$$= \int_0^1 2y\, dy = \left[y^2 \right]_0^1 = 1$$

Thus:

- $f(x, y) \ge 0$
- Total probability = 1

Hence, it is a **valid joint PDF**.

Step 2: Compute $P(A)$

Region:

$$0 \le x \le 0.5, \quad 0.5 \le y \le 1$$

$$P(A) = \int_{0.5}^{1} \int_{0}^{0.5} 4xy \, dx \, dy$$

$$= \int_{0.5}^{1} \left[2x^2 y\right]_0^{0.5} dy = \int_{0.5}^{1} 2y(0.25) \, dy$$

$$= \int_{0.5}^{1} \frac{y}{2} \, dy = \frac{3}{16}$$

---

6. Joint Cumulative Distribution Function (Joint CDF)

Notation

$$X \le x, \; Y \le y \equiv \{X \le x \text{ and } Y \le y\}$$

Definition

$$F(x, y) = P(X \le x, \; Y \le y)$$

---

7. Joint CDF – Continuous Case

If $f(x, y)$ is the joint PDF over $[a, b] \times [c, d]$, then:

$$F(x, y) = \int_{c}^{y} \int_{a}^{x} f(u, v) \, du \, dv$$

Recovering PDF from CDF

$$f(x,y) = \frac{\partial^2}{\partial x \partial y} F(x,y)$$

---

8. Joint CDF – Discrete Case

If $p(x_i, y_j)$ is the joint PMF:

$$F(x,y) = \sum_{x_i \le x} \sum_{y_j \le y} p(x_i, y_j)$$

Interpretation

- Sum probabilities:
  - Rows with $x_i \le x$
  - Columns with $y_j \le y$

---

9. Properties of Joint CDF

The function $F(x,y)$ satisfies:

1. **Non-decreasing property**
   - If x or y increases, $F(x,y)$ does not decrease
2. **Lower-left limit**

$$\lim_{(x,y)\to(-\infty,-\infty)} F(x,y) = 0$$

3. **Upper-right limit**

$$\lim_{(x,y)\to(\infty,\infty)} F(x,y) = 1$$

End of Lecture

If you want, I can convert this into:

- 📄 LaTeX exam notes
- 🧠 1-page revision cheat sheet
- 📊 Important questions + solved numericals